

HOW THE APPLICATION WORKS

CivicAI

Technical Explainer for Team Members

 Single HTML file

 LLM API

 Knowledge Base

 3 Providers

WHAT IS THE APP?

01

From the citizen's perspective

1

Open the file

Double-click public-service-chatbot-final.html in any browser. No install required.

2

Pick a topic

Click a Knowledge Base entry in the sidebar (Permits, Tax, Health...)

3

Ask a question

Type a natural-language question or click a Quick Topic chip.

4

Get an answer

The AI responds in ~2 seconds, grounded in the selected topic's data.

What's inside the file



HTML + CSS

The entire chat interface — sidebar, message bubbles, quick-topic chips, input bar



JavaScript

All application logic: send message, switch topic, call LLM API, render response



Knowledge Base

5 hardcoded topic contexts injected as system prompts into every API call



API Config Block

Three commented provider options (Anthropic / OpenAI / Gemini) — swap with one edit

HOW A MESSAGE IS PROCESSED

02



Citizen types a question

1

User types in the textarea and presses Enter or clicks ►



Active KB context is injected

2

sendMessage() reads KNOWLEDGE_BASE[activeDoc] and prepends it as a system prompt



System prompt is assembled

3

KB context + RULES block are concatenated into one systemPrompt string



fetch() call to LLM API

4

A POST request is sent to the active provider endpoint with the system prompt + user message



LLM generates a grounded answer

5

The model responds using ONLY the knowledge base context — it cannot invent facts outside it



Response rendered in chat

6

mdToHtml() converts markdown → HTML. The bubble appears with a source badge showing the active KB topic

How it works

The KNOWLEDGE_BASE is a JavaScript object at line 440 of the HTML file. Each key is a topic domain. When the user selects a domain in the sidebar, that key's text is injected as the LLM system prompt.

General FAQs

"general"

Office hours, contacts, portal URL, overview of available services. Default on app load.

Building Permits

"permits"

New construction (\$250), renovation (\$80), demolition (\$150). Fees, processing times, required docs, how to apply.

Tax & Payments

"taxes"

Property tax, business license, late penalties, payment methods, senior/low-income relief via Form TR-2.

Health Services

"health"

12 District Health Centers, Emergency Clinic 24/7, free services, insurance accepted, appointments.

Transport & Roads

"transport"

Pothole reporting, 45 bus routes, bus passes, parking permits, signal outages, road closure alerts.

To add a new domain:

Open the HTML file. > find const KNOWLEDGE_BASE. { } > add a new key with your topic text. > add a matching entry in KB_LABELS and a new

THE SYSTEM PROMPT — HOW AI IS CONTROLLED

04

Every API call assembles a system prompt from two parts. This is the key mechanism that prevents hallucination.

PART 1

Knowledge Base Context (dynamic — changes when user switches topic)

```
${KNOWLEDGE_BASE[activeDoc]}
```

→ The full text of the active KB domain (e.g. all permit fees, procedures, contacts)

PART 2

RULES Block (static — same on every call)

1. Answer ONLY from the knowledge base context above
2. Be concise, friendly, and professional
3. Use bullet points for lists of steps or documents
4. If out-of-scope: redirect to +1-800-CIVIC-00 or portal.city.gov
5. NEVER fabricate fees, dates, or procedures

▼ **Result: LLM always stays on-topic and cites only what's in the knowledge base**

SWITCHING LLM PROVIDERS

05

The API config block lives at line 594 of the HTML. Only ONE option should be active at a time. To switch: comment out the active block, uncomment the new one, paste your key.

OPTION A — Anthropic (default, active)

ACTIVE

Key format: sk-ant-...

Model: claude-sonnet-4-6

Response: data.content[0].text

Get at: console.anthropic.com

Auth: Header: x-api-key

OPTION B — OpenAI

Key format: sk-proj-...

Model: gpt-4o / gpt-4o-mini

Response: data.choices[0].message.content

Get at: platform.openai.com/api-keys

Auth: Header: Authorization: Bearer

OPTION C — Google Gemini

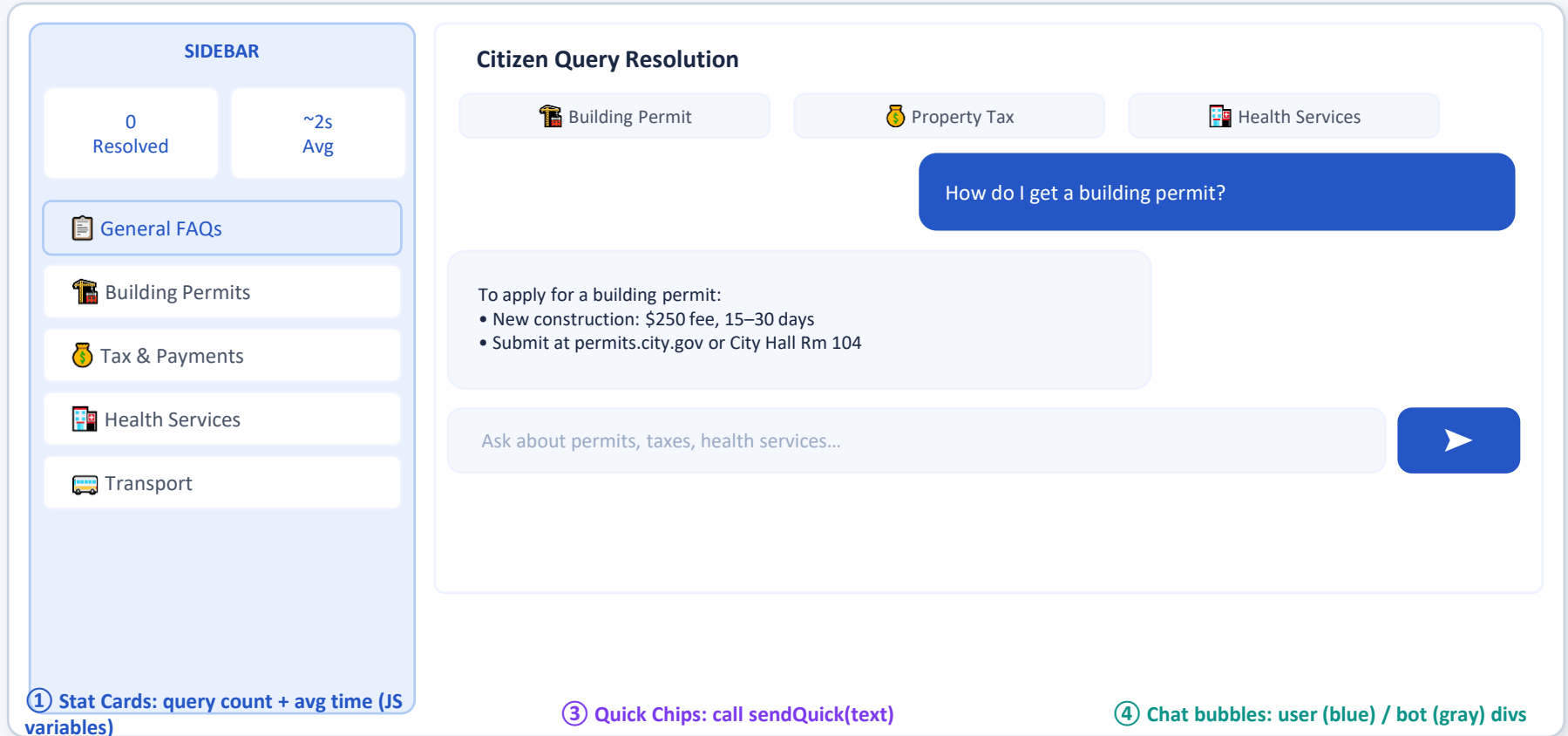
Key format: Alza...

Model: gemini-1.5-flash / gemini-1.5-pro

Response: data.candidates[0].content.parts[0].text

Get at: aistudio.google.com

Auth: Query param: ?key=



CODE STRUCTURE (inside the HTML file)

07

L1 - 280

CSS Styles

All visual design: layout, colors, cards, bubbles, animations, responsive rules. Edit here to change the look.

L281 - 430

HTML Structure

Sidebar (stats + KB list), chat panel (header, chips, messages, input). DOM elements wired to JS by id / onclick.

L431 - 440

State Variables

`let activeDoc = 'general' · let queryCount = 0 · let isProcessing = false`

L440 - 476

KNOWLEDGE_BASE

5 keys: general, permits, taxes, health, transport. Plain text injected as system prompt on every API call.

L477 - 560

Helper Functions

`selectDoc() · autoResize() · handleKey() · sendQuick() · now() · appendMsg() · showTyping() · removeTyping() · mdToHtml()`

L557 - 563

KB_LABELS

Maps key → display name shown in the source badge under each bot response (e.g. 'Building Permits Guide').

L566 - 680

sendMessage() — API Config

Main async function. Assembles system prompt → calls active provider → renders response. API config block at line 594.

HOW TO EXTEND THE KNOWLEDGE BASE

08

To add a new service domain (e.g. 'Waste Management'), make 3 edits in the HTML file:

1

Add the KB entry (around line 440)

```
waste: `Waste Management knowledge base:  
- General waste: collected Mon & Thu. Place bins by 7am.  
- Recycling: blue bin, collected every second Monday.  
- Bulk items: book at waste.city.gov or call 555-WASTE.  
- Hazardous waste: drop-off at 12 Civic Drive, Saturdays 9am-1pm.`
```

2

Add the display label (around line 557)

```
waste: 'Waste Management Guide',
```

3

Add the sidebar button (around line 363 in HTML)

```
<div class="kb-doc" onclick="selectDoc(this, 'waste')">  
  <span class="kb-doc-icon">🗑️</span>  
  <span class="kb-doc-name">Waste Management</span>  
  <span class="kb-doc-tag">GUIDE</span>  
</div>
```

QUICK REFERENCE CARD

09

Keep this handy while working on the code



Open the app

Double-click HTML file in any browser



Switch provider

Line 594 — comment/uncomment Option A/B/C



Paste new API key

Replace the key string in the active Option block



Add KB domain

Slides 8: 3 edits (KB object, KB_LABELS, HTML button)



Change colors

CSS :root variables at top of <style> block



Active model (Anthropic)

claude-sonnet-4-6 (line 620)



Active model (OpenAI)

gpt-4o or gpt-4o-mini (line 634)



Active model (Gemini)

gemini-1.5-flash (line 650)



Edit in VS Code

Ctrl+G → type line number to jump



Add quick chips

HTML ~line 407: copy a .qt-chip div